

Q4 ✓
Digital Certificate Verification
using Set Interface
10 marks

Q5 ✓
Smart Inventory Management
using Map Interface
10 marks

5/5 answered

```

1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner scanner = new Scanner(System.in);
6
7         System.out.print("Enter a paragraph: \n");
8         String originalParagraph = scanner.nextLine();
9
10        System.out.print("Enter the word to replace: \n");
11        String wordToReplace = scanner.nextLine();
12
13        System.out.print("Enter the new word: \n");
14        String replacementWord = scanner.nextLine();
15
16        int origCharCount = originalParagraph.length();
17
18        String[] origWords = originalParagraph.trim().isEmpty() ? new String[0] : originalParagraph.split(" ");
19        int origWordCount = origWords.length;
20
21        String[] origSentences = originalParagraph.trim().isEmpty() ? new String[0] : originalParagraph.split("\\.");
22        int origSentenceCount = origSentences.length;
23
24        StringBuffer editHistory = new StringBuffer();
25        editHistory.append("EDIT HISTORY LOG\n");
26        editHistory.append("-----\n");
27        editHistory.append("Original Document: ").append(originalParagraph);
28        editHistory.append("Character Count: ").append(origCharCount);
29        editHistory.append("Word Count: ").append(origWordCount);
30        editHistory.append("Sentence Count: ").append(origSentenceCount);
31
32        String replacedParagraph = originalParagraph.replace(wordToReplace, replacementWord);
33        editHistory.append("Replaced ").append(wordToReplace).append(" with ").append(replacementWord);
34        editHistory.append("After Replacement: ").append(replacedParagraph);
35
36        String[] replacedSentences = replacedParagraph.split("\\.");
37        StringBuilder finalDocumentBuilder = new StringBuilder();
38
39        for (String sentence : replacedSentences) {
40            StringBuilder sb = new StringBuilder(sentence);
41            finalDocumentBuilder.append(sb.reverse().toString());
42        }
43
44        editHistory.append("Sentences Reversed Individually\n");
45
46        String finalFormattedDocument = finalDocumentBuilder.toString();

```

The cat sat on the mat.
mat
cat

Output

```

Enter a paragraph:
Enter the word to replace:
Enter the new word:
----- ORIGINAL DOCUMENT -----
The cat sat on the mat.

```

10 marks

Q5

Smart Inventory Management
using Map Interface
10 marks

5/5 answered

```

1 import java.util.ArrayList;
2 import java.util.List;
3 import java.util.Scanner;
4
5 public class Main {
6     public static void main(String[] args) {
7
8         List<Patient> patientList = new ArrayList<>();
9
10        patientList.add(new Patient(101, "Alice", "Cardiology", 45));
11        patientList.add(new Patient(102, "Bob", "Neurology", 30));
12        patientList.add(new Patient(103, "Charlie", "Cardiology", 50));
13        patientList.add(new Patient(104, "David", "Orthopedics", 28));
14
15        System.out.println("--- Registered Patients (Registration Order) ---");
16        for (Patient p : patientList) {
17            System.out.println(p);
18        }
19
20        int updateId = 102;
21        for (Patient p : patientList) {
22            if (p.id == updateId) {
23                p.name = "Bob Smith";
24                p.age = 32;
25                break;
26            }
27        }
28
29        int cancelId = 103;
30        patientList.removeIf(p -> p.id == cancelId);
31
32        String targetDept = "Cardiology";
33        System.out.println("\n--- Patients in " + targetDept + " Department ---");
34        for (Patient p : patientList) {
35            if (p.department.equalsIgnoreCase(targetDept)) {
36                System.out.println(p);
37            }
38        }
39
40        System.out.println("\n--- First and Last Registered Patients ---");
41        if (!patientList.isEmpty()) {
42            System.out.println("First Registered Patient: " + patientList.get(0));
43            System.out.println("Last Registered Patient: " + patientList.get(patientList.size() - 1));
44        }
45    }

```

stdin input

Output

```

--- Registered Patients
(Registration Order) ---
ID: 101
Name: Alice
Department: Cardiology
Age: 45

```

Original question is not available
using Set Interface
10 marks

Q5 
Smart Inventory Management
using Map Interface
10 marks

5/5 answered

```

1 import java.util.*;
2
3 public class AirportBoardingQueue {
4
5     public static void main(String[] args) {
6         Queue<String> regularQueue = new LinkedList<>();
7         PriorityQueue<VIPPassenger> vipQueue = new PriorityQueue<>();
8
9         regularQueue.add("Arun");
10        regularQueue.add("Bala");
11        regularQueue.add("Chitra");
12        regularQueue.add("Deepa");
13
14        vipQueue.add(new VIPPassenger("Elite Karthik", 1));
15        vipQueue.add(new VIPPassenger("Gold Meena", 3));
16        vipQueue.add(new VIPPassenger("Platinum Suresh", 2));
17        System.out.println("==== Airport Boarding Queue Management ===");
18
19        System.out.println("Current Regular Passenger Queue: " + regularQueue);
20        System.out.println("Current VIP Passenger Queue: " + vipQueue);
21        System.out.println("\n----- Boarding VIP Passengers First (Priority)");
22        while (!vipQueue.isEmpty()) {
23            VIPPassenger nextVIP = vipQueue.peek();
24            System.out.println("Next VIP to board (peek): " + nextVIP);
25            VIPPassenger boardedVIP = vipQueue.poll();
26            System.out.println("Boarded: " + boardedVIP);
27
28            System.out.println("Remaining VIP Queue: " + vipQueue + "\n");
29        }
30
31        System.out.println("----- Boarding Regular Passengers (FIFO Order)");
32        while (!regularQueue.isEmpty()) {
33            String nextPassenger = regularQueue.peek();
34            System.out.println("Next passenger to board (peek): " + nextPassenger);
35
36            String boardedPassenger = regularQueue.poll();
37            System.out.println("Boarded: " + boardedPassenger);
38
39            System.out.println("Remaining Regular Queue: " + regularQueue + "\n");
40        }
41        System.out.println("==== All Passengers Boarded Successfully =====");
42    }
43 }
44
45 class VIPPassenger implements Comparable<VIPPassenger> {
46     String name;

```

stdin input

Output

```

==== Airport Boarding Queue
Management ====

Current Regular Passenger Queue:
[Arun, Bala, Chitra, Deepa]
Current VIP Passenger Queue:

```

using Set Interface
10 marks

Q5 ✓
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5/5 answered

Your Code

```

1 import java.util.*;
2
3 public class Main {
4     public static void main(String[] args) {
5
6         HashSet<String> certificates = new HashSet<>();
7
8         certificates.add("CERT101");
9         certificates.add("CERT102");
10        certificates.add("CERT103");
11        certificates.add("CERT104");
12        certificates.add("CERT102");
13
14        System.out.println("Certificates (Insertion Order):");
15        System.out.println(certificates);
16
17        String search = "CERT103";
18        if (certificates.contains(search))
19            System.out.println(search + " Found");
20        else
21            System.out.println(search + " Not Found");
22
23        TreeSet<String> sorted = new TreeSet<>(certificates);
24
25        System.out.println("Certificates (Sorted Order):");
26        System.out.println(sorted);
27
28        System.out.println("Total Unique Certificates: " + certificates.size());
29
30        certificates.remove("CERT104");
31
32        System.out.println("After Removing Revoked Certificate:");
33        System.out.println(certificates);
34    }
35 }

```



Run



Save

Input

stdin input

Output

```

Certificates (Insertion Order):
[CERT101, CERT102, CERT103,
CERT104]
CERT103 Found
Certificates (Sorted Order):
[CERT101, CERT102, CERT103]

```


Q5 

Smart Inventory Management
using Map Interface
10 marks

5/5 answered

```

1 import java.util.*;
2
3 public class Product {
4
5     String name, category;
6     int quantity;
7     double price;
8
9     Product(String name, String category, int quantity, double price) {
10         this.name = name;
11         this.category = category;
12         this.quantity = quantity;
13         this.price = price;
14     }
15
16     public static void main(String[] args) {
17
18         HashMap<Integer, Product> map = new HashMap<>();
19
20         map.put(101, new Product("Laptop", "Electronics", 5, 50000));
21         map.put(102, new Product("Mobile", "Electronics", 10, 15000));
22         map.put(103, new Product("Book", "Education", 20, 300));
23
24         map.get(102).quantity = 15;
25
26         int id = 102;
27
28         if (map.containsKey(id))
29             System.out.println("Product Found: " + map.get(id).name);
30         else
31             System.out.println("Product Not Found");
32
33         double total = 0;
34
35         for (Product p : map.values())
36             total += p.quantity * p.price;
37
38         System.out.println("Total Inventory Value: " + total);
39
40         System.out.println("Category-wise Products:");
41
42         for (Map.Entry<Integer, Product> e : map.entrySet()) {
43             Product p = e.getValue();
44             System.out.println(p.category + " : " + p.name);
45         }

```

stdin input

Output

```

Product Found: Mobile
Total Inventory Value: 481000.0
Category-wise Products:
Electronics : Laptop
Electronics : Mobile
Education : Book

```